

Name: A Proletariat(Joy, Reynie and Emerson)

Date: 03/10/2026 **Class:** Period 6 Engineering Systems

MyDesign Portfolio

Element B: Electronic Panel Cover

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design A - Clipboard Protection System

Source: Design: Ms. Anderson

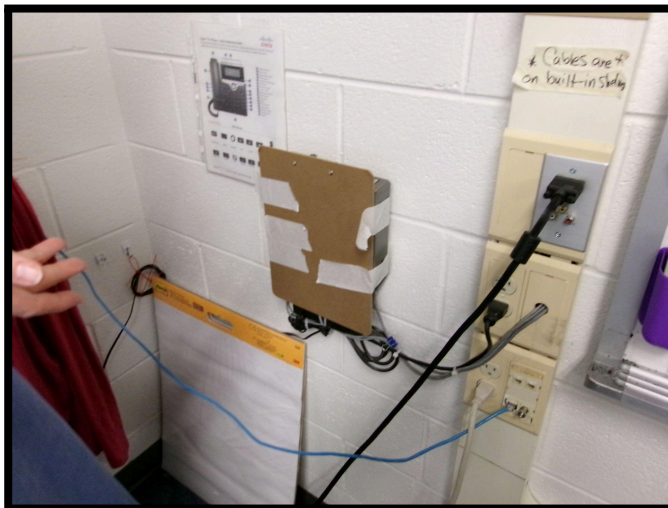


Image:

Details: The clipboard is attached using white painter's tape as well being placed above the control center for the audio box. The clipboard extends down just far enough to block the chair from wedging itself under the audio box. Instead, the chair hits the clipboard. Attachment is simple, but requires continuous disassembly for access. It also does not solve the problem of cable management.

Analysis: The clipboard was used to block the chair from wedging itself in between the floor and the AUX cable box, however it does not look aesthetically pleasing and additionally requires removal and replacement every time she needs to use a clipboard. Therefore making the solution inviable.

Prior Design Attempt #2 - Cable Protection Box

Source: "SOCKiTBOX Components - SOCKiTBOX." SOCKiTBOX - SOCKiTBOX - Connections Made Simple, 17 Jan. 2013, www.sockitbox.com/design/dribox-components/. Accessed 5 June 2026.

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Image:

Details: This is an existing solution to protecting cable connections. Our stakeholder had an issue where the jack of the aux cable would cleanly snap off, so this would prevent that issue by directly protecting the pins. It has a locking clamp and is made for extension cords, but would function for the audio box as well.

Patent Certifications:

Certification - SOCKiTBX. (2013, January 17). SOCKiTBX - SOCKiTBX - Connections Made Simple. <https://www.sockitbox.com/design/certification/>

The Patent for this product exists, but the company has hidden the text, with the only available segments being the text from the US Patent Search and the Patent Citation.

Available Patent Text:

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Cable Container

DOCUME DATE
NT ID PUBLISH
US
D108912 2025-08-
1 S 19

INVENTOR INFORMATION

NAME	CITY	STAT E	ZIP COD E	COU NTRY
Ruddick; Paul	Tyne & Wear	N/A	N/A	GB
Bianchi; Stephen	Tyne & Wear	N/A	N/A	GB

ASSIGNEE INFORMATION

NAME	CITY	STAT E	ZIP COD E	COU NTRY
D-Line (Europe) Limited	Tyne & Wear	N/A	N/A	GB

TYPE CODE
03

APPLICATION NO **DATE FILED**
D/861347 2022-11-29

FOREIGN APPLICATION PRIORITY DATA

COUNTRY	APPLICATION N NO	APPLICATION N DATE
GB	6211759	2022-06-01
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GB	6211762	2022-06-01
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GB	6211764	2022-06-01

US CLASS CURRENT:

D13/149

Description

- (1) FIG. 1 is a top perspective view of a cable container showing our new design.
(2) FIG. 2 is a bottom perspective view thereof.
(3) FIG. 3 is a top view thereof.
(4) FIG. 4 is a bottom view thereof.
(5) FIG. 5 is a left side view thereof.
(6) FIG. 6 is a right side view thereof.
(7) FIG. 7 is a back view thereof; and,

(8) FIG. 8 is a front view thereof.

Claims

The ornamental design for a cable container as shown and described.

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Ruddick, P., Bianchi, S.(2025). *Cable Container*(U.S. Patent No. 1,089,121). U.S. Patent and Trade-mark Office.

Analysis: This socket-lution has a prohibitively high cost (\$32) and is not made by the designers but rather an outside factor. Additionally, our stakeholder wanted the solution to be low profile and blend in, so this blockier design and its bright green color wouldn't meet that requirement.

Prior Design Attempt #3 - Cord raceway

Source: [MonoSystems CordHider 60-in L White Raceway CR1-W at Lowes.com](https://www.monosystems.com/wp-content/themes/monosystems/temp_archives/catalog_pdfs/FWH-FCL-CableHider_PVC_Raceway.pdf)

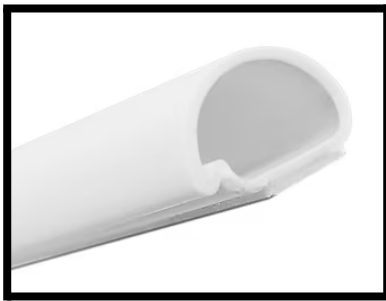


Image:

Details: This is a very simple path made out of a flexible material that is used to line walls. It is thick enough to stop the chair from wedging into the back wall and it also flattens the cables against the wall. It can be lined all the way up to the plugin point, meaning the pin would be protected.

Patent and Information:

https://www.monosystems.com/wp-content/themes/monosystems/temp_archives/catalog_pdfs/FWH-FCL-CableHider_PVC_Raceway.pdf

Handler, J. (2018). *Articulating On-floor Electric Power Strip Including A Cord Protector And Cord Outlets* (USPTO Patent).

MonoSystems. (n.d.). *Raceway Fittings FCA-150 Surface Raceway / Non-metallic Notes*.

Retrieved June 15, 2026, from

https://www.monosystems.com/wp-content/themes/monosystems/temp_archives/catalog_pdfs/FWH-FCL-CableHider_PVC_Raceway.pdf

Analysis: This solution would make the cables less visible, tying into our stakeholder's goal of a low profile design. However, this wouldn't prevent the chair from rolling under the box, nor preventing the cable from breaking.

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Prior Design Attempt #4 - Right Angled Aux Cable

Source: [3.5mm Headphone Extension Cable 6FT Right Angle 3.5mm Male to Female Stereo Audio Extension Cable/90 Degree Right Angle Aux Cable for Smartphones, Tablets, Media Players](#)



Image:

Details: This would keep the wire at an angle, allowing our stakeholder to more comfortably roll her chair under the aux cable box. Angling the cable backwards would make the cable stronger and prevent the cable from breaking as easily.

Analysis: While the cord would make the actually risk of having the jack break lower, the core problem of having the chair wedge itself under the AUX cable box still isn't fixed and therefore, while each instance of the chair getting wedged would be less likely to cause problems, it will become just as problematic over a slightly longer timeframe.

Stepancich, G., & Shenkal, Y. (2002). *Retractable cable assembly* (USPTO Patent).
<https://patents.google.com/patent/US20020040945A1/en>

Prior Design Attempt #5 - Charger Cable Saver Protector Spiral Tube Wire Management Organizer Protective Cord Sleeve

Source:

<https://www.amazon.com/KDDOM-Protector-Management-Organizer-Protective/dp/B08LSTJL19>

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Image :

Details: The cord sleeve would prevent the transition between the cord and the jack to keep the cord from becoming frayed. It would also increase the ability of the cord to withstand force from other sources. And would look blended in, which helps the aesthetic goals we have.

Analysis: While the cord sleeve can protect certain parts of the cord from being damaged, the sleeve cannot protect the actual jack from breaking off when the chair wedges itself underneath the AUX cable box as the sleeve is in no way adding any form of durability to the weak point of the cable.

Carnevali, J. (2024). *Docking sleeve with electrical adapter Abstract* (USPTO Patent).
<https://patents.google.com/patent/US12143141B2/en>